

# Engineering Project Portfolio

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*Electrical & Electronics Engineer*



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## About Me

I am an Electrical & Electronics Engineer and Master of Aerospace Engineering candidate at the University of Auckland with interests in spacecraft systems, embedded systems, RF communications, and system design. I enjoy tackling engineering problems that require balancing technical constraints and developing practical, well-reasoned solutions through systems thinking.

Throughout my studies and professional experience, I have worked on projects spanning embedded systems, PCB design, robotics, simulation, aerospace systems, and software development. These opportunities have strengthened both my technical knowledge and my ability to evaluate engineering trade-offs, integrate multidisciplinary systems, and approach complex problems with a structured mindset.

I value collaboration, continuous learning, and clear communication. Engineering competitions, international projects, and technical presentations have reinforced the importance of working effectively with others and communicating complex ideas in a way that is accessible to different audiences.

As I continue my Master's research, I hope to build a career in the space industry where I can contribute to the design, development, and operation of advanced aerospace systems. I am particularly motivated by projects that combine research, innovation, and systems engineering to solve challenging real-world problems and support the future of space exploration.

[Insert self image]

## Skills Overview

### Systems Engineering

|                            |                            |
|----------------------------|----------------------------|
| Systems Architecture<br>●○ | Requirements Analysis      |
| Trade Studies              | Verification & Validation  |
| Systems Integration        | Mission Analysis           |
| Operational Analysis       | Engineering Design Process |

### Aerospace Engineering

|                        |                                |
|------------------------|--------------------------------|
| Spacecraft Systems     | Orbital Mechanics              |
| RF Communications      | Link Budget Analysis           |
| Mission Design         | Space Systems Engineering      |
| Thermal Considerations | Launch Operations Fundamentals |

### Programming & Software Development

|              |              |
|--------------|--------------|
| Python       | MATLAB       |
| C            | C++          |
| C#           | Arduino      |
| Raspberry Pi | Git / GitHub |

### Electrical & Electronics Engineering

|                               |                               |
|-------------------------------|-------------------------------|
| Electrical System Design      | Embedded Systems Development  |
| Electrical Power Distribution | Hardware-Software Integration |
| Wiring Harness Assembly       | PCB Design                    |
| Electrical System Integration | Electronics Prototyping       |
| Power Electronics             | Soldering & PCBA Rework       |

|                               |                         |
|-------------------------------|-------------------------|
| Analogue Electronics          | Digital Electronics     |
| Circuit Design & Analysis     | Circuit Debugging       |
| Signal Conditioning           | Microcontrollers        |
| Instrumentation & Measurement | Electronics Diagnostics |

#### CAD, Simulation & Engineering Tools

|                 |            |
|-----------------|------------|
| SolidWorks      | Fusion 360 |
| Altium Designer | Multisim   |
| LabVIEW         | SUMO       |
| Visual Studio   | VS Code    |

#### Testing & Validation

|                           |                         |
|---------------------------|-------------------------|
| Functional Testing        | Fault Finding           |
| Electronics Diagnostics   | Verification Testing    |
| Oscilloscope Measurements | Multimeter Measurements |
| Hardware Debugging        | System Validation       |

#### Technical Documentation

|                      |                            |
|----------------------|----------------------------|
| Technical Reports    | Engineering Drawings       |
| Test Documentation   | Requirements Documentation |
| Design Documentation | Technical Presentations    |

#### Professional Skills

|                         |                                |
|-------------------------|--------------------------------|
| Problem Solving         | Critical Thinking              |
| Technical Communication | Cross-functional Collaboration |
| Teamwork                | Attention to Detail            |
| Continuous Learning     | Project Planning               |

## Projects

### Space-Based AI Data Centre System Concept

#### Project Summary

|              |                                                           |
|--------------|-----------------------------------------------------------|
| Project Type | University Systems Engineering Project                    |
| Status       | Complete                                                  |
| Institution  | University of Auckland                                    |
| Discipline   | Aerospace Systems Engineering                             |
| Role         | Systems Engineering Team Member & Architect               |
| Focus Areas  | Mission Architecture, Systems Engineering, Programme Plan |

|                  |                                                                                                                                                                                                                             |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Key Contribution | Led technical meetings, authored major systems engineering sections and modelled system.                                                                                                                                    |
| Tools Used       | <ul style="list-style-type: none"> <li>• Google Docs</li> <li>• Google Sheets</li> <li>• Google Meet</li> <li>• Draw.io</li> <li>• Systems Engineering Methodologies</li> <li>• Technical Research</li> <li>• AI</li> </ul> |
| Duration         | Semester 1, 2026                                                                                                                                                                                                            |
| Team Size        | 8                                                                                                                                                                                                                           |

## Overview

A multidisciplinary systems engineering project investigating the feasibility of a space-based AI data centre capable of supporting future computational demands while reducing reliance on terrestrial infrastructure. The project evaluated mission architecture, spacecraft systems, lifecycle planning, financial feasibility, and operational sustainability to produce a complete mission concept.

## Objectives

- Research and develop a technically feasible concept for a space-based AI data centre.
- Design an end-to-end spacecraft system architecture.
- Evaluate operational, maintenance, and end-of-life strategies.
- Produce a realistic programme plan and financial assessment.
- Design and model system operations and interfaces
- Apply systems engineering principles throughout the project lifecycle.

## Solution Overview

Following a series of systems engineering trade studies, our team selected a **Hybrid LEO–GEO architecture** as the preferred mission concept. The design separated AI model training and inference into different orbital segments to take advantage of the strengths of each orbit. A dedicated **GEO platform** was proposed for computationally intensive AI model training and large-scale data processing, as its continuous visibility to ground stations and stable orbital environment make it well suited for long-duration workloads and frequent data exchange.

A **distributed constellation of LEO nodes** was designed to provide low-latency AI inference services closer to Earth, improving responsiveness while allowing the system to scale through multiple interconnected satellites. Communication between the GEO platform and LEO constellation was achieved through a ground-mediated network that transferred training datasets, model updates, and inference data. This hybrid approach provided a practical balance between computational capability, operational efficiency, scalability, and long-term sustainability, making it the preferred solution over the alternative LEO-only, MEO, and GEO-only architectures considered during the project.

[add architecture image]

Engineering Process

The project followed a structured **systems engineering methodology**, using the **Systems Engineering V-Model** to guide development from mission definition through to the preliminary system design. Mission objectives, stakeholder requirements, and operational needs were progressively translated into system requirements, subsystem architectures, and verification planning, ensuring traceability throughout the design process. The preliminary design phase was managed using a **Waterfall development approach**, allowing each stage to build systematically on the completion of the previous one.

The project progressed through mission analysis, requirements definition, concept generation, architecture development, subsystem design, trade studies, lifecycle planning, and programme management before producing the final conceptual design. Trade studies evaluated alternative deployment, communication, and compute architectures against performance, cost, scalability, sustainability, and operational feasibility, ultimately leading to the selection of a **Hybrid LEO–GEO architecture** supported by a **Protected COTS compute architecture** as the preferred baseline solution.

[add system engineering process diagram]

My Contributions

I played a leading role in both the technical development and coordination of the project. My primary responsibilities included authoring and developing the following areas:

|                       |                                                                                                         |
|-----------------------|---------------------------------------------------------------------------------------------------------|
| Mission Overview      | Defined the mission purpose, objectives, and overall project scope.                                     |
| Concept of Operations | Developed the operational concept describing how the system functions throughout the mission lifecycle. |
| Requirements Analysis | Translated stakeholder needs into system-level requirements to guide the design process.                |
| System Overview       | Produced a high-level overview of the proposed system and its major functional elements.                |
| Trade Studies         | Evaluated alternative architectures and design options to support engineering decision-making.          |
| Systems Modelling     | Modelled subsystem interactions and operational workflows within the proposed architecture.             |
| System Architecture   | Designed the overall system architecture, defining subsystem integration and interfaces.                |
| Maintenance Strategy  | Developed the maintenance approach to support long-term operational reliability.                        |
| Upgrade Strategy      | Planned system upgrade pathways to improve scalability and future adaptability.                         |
| Disposal Strategy     | Defined the end-of-life disposal approach to support sustainable orbital operations.                    |
| Programme Plan        | Produced the programme schedule, milestones, and implementation roadmap.                                |

|                                    |                                                                                                                                                 |
|------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| Life Cycle Planning                | Developed the system lifecycle plan, defining activities from deployment and operation through maintenance, upgrades, and end-of-life disposal. |
| System Budgets                     | Developed engineering budgets for mass, power, communications, and other system resources to support the proposed mission architecture.         |
| Financial Budget and Cost Planning | Prepared programme cost estimates and financial planning to assess project feasibility.                                                         |

In addition to my technical responsibilities, I led many of the team's design meetings, facilitating technical discussions, coordinating decision-making, and helping the team converge on the final system architecture through collaborative engineering trade-offs.

### Skills Demonstrated

|                            |                                                                                                    |
|----------------------------|----------------------------------------------------------------------------------------------------|
| Requirements Analysis      | Translated stakeholder needs into system requirements that guided the mission architecture.        |
| Systems Engineering        | Applied the Systems Engineering V-Model to progress from mission definition to preliminary design. |
| Systems Modelling          | Modelled subsystem interactions, operational concepts, and lifecycle strategies.                   |
| Engineering Trade Studies  | Evaluated architectural concepts and deployment strategies to select the preferred solution.       |
| Space Mission Architecture | Developed the Hybrid LEO–GEO architecture separating AI training and inference.                    |
| Spacecraft Systems Design  | Defined subsystem architecture, integration, and operational concepts.                             |
| Lifecycle Engineering      | Developed maintenance, upgrade, and disposal strategies for long-term sustainability.              |
| Mission Planning           | Defined mission objectives, deployment strategy, and operational roadmap.                          |
| Project Management         | Produced project schedules, milestones, and implementation plans.                                  |
| Cost & Budget Analysis     | Estimated programme costs and assessed financial feasibility.                                      |
| Technical Documentation    | Authored system architecture, lifecycle, programme planning, and financial documentation.          |
| Technical Leadership       | Led technical meetings and facilitated engineering decision-making.                                |
| Team Collaboration         | Worked within a multidisciplinary team to develop the final mission concept.                       |

### Challenges

One of the main challenges was balancing ambitious mission objectives with realistic engineering constraints. Designing a technically feasible system while considering maintenance, upgradeability, disposal, programme scheduling, and financial limitations required continuous collaboration and iterative design refinement. Leading technical discussions also involved integrating diverse viewpoints into a coherent systems solution.

Results

The project produced a comprehensive mission concept for a space-based AI data centre, supported by detailed systems engineering documentation, lifecycle planning, programme management, and financial analysis. The final design demonstrated the application of systems engineering principles to a complex aerospace concept from mission conception through end-of-life planning. This led to achieving the highest grade in the cohort.

[add spacecraft configuration image]

[add operational life cycle image]

Lessons Learned

This project strengthened my understanding of large-scale systems engineering by demonstrating how technical, operational, financial, and programme management considerations must all be integrated into a successful mission concept. It also developed my confidence in leading engineering discussions, coordinating multidisciplinary teams, and communicating complex technical ideas.

Images / Diagrams

Supporting Material

J.A.R.V.I.S Voice Controller

Project Summary

|                  |                                                                                                                                                                                                   |
|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Project Type     | Independent Engineering Project                                                                                                                                                                   |
| Status           | Prototype Development (Ongoing)                                                                                                                                                                   |
| Institution      | Independent                                                                                                                                                                                       |
| Discipline       | Embedded Systems, Internet of Things, Artificial Intelligence                                                                                                                                     |
| Role             | Sole Developer                                                                                                                                                                                    |
| Focus Areas      | Voice Recognition, Embedded Systems, Hardware–Software Integration, AI Integration, IoT                                                                                                           |
| Key Contribution | Designed and developed the complete system architecture, implemented the Python–Arduino communication framework, integrated voice command functionality, and built the initial working prototype. |
| Tools Used       | <ul style="list-style-type: none"><li>Python</li></ul>                                                                                                                                            |



|           |                                                                                                                   |
|-----------|-------------------------------------------------------------------------------------------------------------------|
|           | <ul style="list-style-type: none"> <li>• Arduino</li> <li>• Raspberry Pi</li> <li>• Visual Studio Code</li> </ul> |
| Duration  | 2026 - Present                                                                                                    |
| Team Size | Independent                                                                                                       |

## Overview

Inspired by **J.A.R.V.I.S.**, the iconic virtual assistant from the *Iron Man* films, this independent engineering project aims to develop a modular voice-controlled assistant capable of interacting with embedded hardware. While fictional, J.A.R.V.I.S. has inspired engineers and technology enthusiasts by showcasing the potential of intelligent human–computer interaction, motivating the development of systems that bridge software and the physical world.

The long-term vision is to combine speech recognition, artificial intelligence, microcontrollers, and IoT technologies into a scalable platform capable of controlling real-world devices through natural language. The current prototype focuses on establishing a reliable communication framework between Python and an Arduino microcontroller, validating the core hardware–software architecture that will support future integration of voice recognition, large language models, computer vision, wireless connectivity, and intelligent automation.

## Objectives

- Design and develop a modular AI-powered voice assistant capable of interacting with embedded hardware through natural language commands.
- Establish reliable communication between Python and Arduino to enable real-time control of physical devices.
- Develop a scalable system architecture that supports future integration of additional hardware, sensors, and intelligent software modules.
- Investigate the practical application of artificial intelligence, embedded systems, and IoT technologies within a unified engineering platform.
- Apply iterative prototyping and testing to validate hardware–software integration and improve system reliability.
- Build a long-term engineering platform that can evolve to incorporate advanced capabilities such as computer vision, large language models, autonomous decision-making, and smart automation.

## Solution Overview

The current prototype establishes the core communication framework between a Python application and an Arduino microcontroller through serial communication. User commands entered the Python interface are transmitted to the Arduino, where they are interpreted and executed to control connected electronic components on a breadboard. This validates the underlying hardware–software communication pipeline that forms the foundation of the project.

At its current stage, the system successfully executes commands such as **"light on"**, **"light off"**, **"motor on"**, and **"motor off"**, allowing LEDs and a DC motor to be controlled through a simple

command interface. The modular architecture has been designed to support future expansion, with planned developments including voice recognition, large language model integration, computer vision, Raspberry Pi deployment, wireless IoT connectivity, and support for additional sensors and actuators as the project evolves.

[add architecture image]

Engineering Process

The development of **JARVIS** has followed an iterative prototyping approach, with the initial objective of establishing reliable communication between software and embedded hardware before introducing more advanced functionality. The project began by defining the system requirements and selecting appropriate hardware and software platforms. An Arduino Uno was chosen as the embedded controller, while Python was selected as the primary interface for communicating with the microcontroller via serial communication over USB, providing a simple and reliable platform for validating the system architecture.

Following hardware assembly on a breadboard, Arduino firmware was developed to interpret predefined serial commands, while a Python application was implemented to transmit commands and control connected devices. Functional testing was carried out to verify reliable communication, command execution, and overall system stability using LEDs and a DC motor. The project continues to evolve through incremental development, with future iterations planned to incorporate voice recognition, computer vision, large language models, wireless connectivity, and broader IoT capabilities.

My Contributions

As the sole developer of **JARVIS**, I have been responsible for the complete design, development, and testing of the prototype from concept through implementation. The project has provided an opportunity to apply embedded systems engineering principles while gaining practical experience in hardware–software integration.

My key responsibilities include:

|                               |                                                                                                                                        |
|-------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| System Concept & Architecture | Designed the overall system concept and a modular architecture to support future expansion.                                            |
| Prototype Development         | Designed and assembled the breadboard prototype, integrating LEDs, a DC motor, and supporting circuitry.                               |
| Embedded Systems Programming  | Developed the Arduino firmware to receive and execute serial commands for hardware control.                                            |
| Software Development          | Developed the Python application to establish serial communication and transmit user commands.                                         |
| Hardware–Software Integration | Integrating the hardware and software components into a functional prototype.                                                          |
| Testing & Debugging           | Performed functional testing and debugging to validate reliable communication and device operation.                                    |
| Future Development Planning   | Defined the project roadmap for future integration of voice recognition, large language models, computer vision, and IoT connectivity. |

Developing JARVIS has allowed me to gain practical experience across multiple engineering disciplines, including embedded systems, electronics, programming, systems integration, and iterative prototyping.

### Skills Demonstrated

|                               |                                                                                                   |
|-------------------------------|---------------------------------------------------------------------------------------------------|
| Systems Architecture          | Designed a modular hardware–software architecture to support future expansion.                    |
| Embedded Systems              | Developed Arduino firmware to control electronic devices through serial commands.                 |
| Python Programming            | Developed the Python application for serial communication and command transmission.               |
| Hardware–Software Integration | Integrated Python and Arduino into a functional embedded control system.                          |
| Serial Communication          | Implemented USB serial communication between the PC and Arduino.                                  |
| Electronics Prototyping       | Designed and assembled the breadboard prototype using LEDs, a DC motor, and supporting circuitry. |
| Circuit Integration           | Integrated electronic components into a functional prototype for testing.                         |
| Functional Testing            | Verified reliable command execution and hardware operation through iterative testing.             |
| Debugging & Troubleshooting   | Identified and resolved software and hardware integration issues during development.              |
| Technical Documentation       | Documented the system architecture, prototype development, and future roadmap.                    |
| Engineering Design            | Applied an iterative prototyping approach to validate the system before future development.       |
| Project Planning              | Defined the long-term development roadmap for future intelligent system capabilities.             |

### Challenges

One of the primary challenges was establishing reliable communication between the Python application and the Arduino microcontroller while ensuring commands were transmitted and executed consistently. Developing and debugging the serial communication interface required iterative testing to verify correct command interpretation, hardware response, and overall system stability before additional functionality could be introduced.

Another challenge was designing the prototype with future expansion in mind. Rather than developing a single-purpose controller, the system architecture was designed to remain modular so that features such as voice recognition, large language models, computer vision, and IoT connectivity can be integrated without requiring a complete redesign of the existing hardware–software framework.

Results

The project successfully produced a functional proof-of-concept prototype demonstrating reliable communication between a Python application and an Arduino microcontroller. The system was able to receive predefined user commands and control physical hardware, such as the LEDs and a DC motor, validating the core hardware–software communication architecture.

The successful prototype provides a solid foundation for future development and demonstrates the feasibility of expanding the system into a more capable intelligent assistant. By validating the underlying communication framework at an early stage, the project has established a scalable platform that can support future integration of voice recognition, artificial intelligence, computer vision, and IoT technologies.

[Insert prototype image]

Lessons Learned

Developing JARVIS reinforced the importance of building reliable foundations before introducing additional complexity. By focusing on hardware–software communication first, I gained a greater appreciation for validating individual subsystems through iterative prototyping before expanding overall system functionality.

The project also strengthened my understanding of embedded systems integration, modular software architecture, and the value of designing with future scalability in mind. It highlighted how thoughtful system design during the early stages can simplify future development and support the integration of more advanced capabilities as the project evolves.

Images / Diagrams

Supporting Material

*In Development*

Aircraft & Atmosphere System Concept

Project Summary

|                  |                                                                                                                                                                 |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Project Type     | University Systems Engineering Project                                                                                                                          |
| Status           | Complete                                                                                                                                                        |
| Institution      | University of Auckland                                                                                                                                          |
| Discipline       | Aeronautical Systems Engineering                                                                                                                                |
| Role             | Systems Engineering Team Member & Architect                                                                                                                     |
| Focus Areas      | Aircraft Systems Design, Mission Analysis, Systems Engineering, Subsystem Integration                                                                           |
| Key Contribution | Authored the Mission Overview, System Overview, subsystem architecture models, Electrical & Environmental Control System, and Avionics subsystem documentation. |

|            |                                                                                                                                                                                                                             |
|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Tools Used | <ul style="list-style-type: none"> <li>• Google Docs</li> <li>• Google Sheets</li> <li>• Google Meet</li> <li>• Draw.io</li> <li>• Systems Engineering Methodologies</li> <li>• Technical Research</li> <li>• AI</li> </ul> |
| Duration   | Semester 1, 2026                                                                                                                                                                                                            |
| Team Size  | 6                                                                                                                                                                                                                           |

## Overview

This university systems engineering project involved the conceptual design of a civilian maritime surveillance and search-and-rescue aircraft to complement New Zealand's existing P-8A Poseidon fleet. Working within operational, regulatory, and cost constraints, the project applied a structured systems engineering approach to develop a preliminary aircraft capable of conducting maritime surveillance, search and rescue, and aid delivery across New Zealand's extensive maritime area of interest.

The project progressed from mission definition and stakeholder analysis through requirements development, system architecture, subsystem integration, and preliminary aircraft design. Based on trade studies, the Pilatus PC-12 PRO Special Mission platform was selected as the preferred solution, integrating surveillance sensors, avionics, communication and navigation systems, and search-and-rescue capabilities into a practical, cost-effective aircraft concept that balanced operational performance, maintainability, and certification requirements.

## Objectives

- Develop a conceptual civilian maritime surveillance aircraft to complement New Zealand's existing maritime patrol capability.
- Apply a structured systems engineering approach to define mission requirements, system architecture, and subsystem integration.
- Design an aircraft capable of conducting maritime surveillance, search and rescue, and emergency aid delivery while operating from regional airfields.
- Develop and integrate the aircraft's major subsystems, including avionics, electrical, environmental control, communications, and surveillance systems, into a cohesive preliminary design.
- Evaluate design decisions against operational performance, certification requirements, maintainability, lifecycle considerations, and programme cost to produce a practical and feasible mission solution.

## Solution Overview

Following a series of systems engineering trade studies, the team selected the Pilatus PC-12 PRO Special Mission as the preferred platform for a civilian maritime surveillance and search-and-rescue aircraft. Rather than developing a new airframe, the solution leveraged an existing certified aircraft and integrated mission-specific capabilities, including maritime radar, EO/IR

sensors, AIS reception, communications and navigation systems, mission management equipment, and emergency payload deployment. This approach reduced development and certification risk while delivering a practical, cost-effective capability for New Zealand's maritime operations.

The proposed system was designed to conduct long-range maritime surveillance, identify vessels during day and night operations, coordinate with rescue agencies, and deploy life-saving equipment when required. The PC-12 was selected because it offered the best balance of range, short-field performance, operating cost, maintainability, and mission-system integration compared with alternative aircraft considered during the trade studies, providing a scalable solution that complements, rather than replaces the RNZAF P-8A Poseidon fleet.

[insert Aircraft layout]

Engineering Process

The project followed a structured systems engineering approach, progressing from mission definition and stakeholder analysis through requirements development, concept generation, trade studies, subsystem design, and verification planning. Mission objectives were translated into system-level requirements before being decomposed into functional architectures and integrated aircraft subsystems, ensuring traceability throughout the preliminary design process.

Trade studies were undertaken to evaluate candidate aircraft platforms against operational capability, cost, maintainability, certification requirements, and mission suitability, leading to the selection of the Pilatus PC-12 PRO Special Mission as the preferred baseline. The preliminary design was then refined through subsystem architecture development, interface modelling, and verification planning to ensure the proposed aircraft satisfied the defined mission and operational requirements.

My Contributions

|                                               |                                                                                                                  |
|-----------------------------------------------|------------------------------------------------------------------------------------------------------------------|
| Mission Overview                              | Developed the mission overview, including project scope, objectives, stakeholders, ConOps, and success criteria. |
| Requirements Analysis                         | Defined the top-level system requirements from mission and stakeholder needs.                                    |
| System Overview                               | Produced the overall system overview and high-level aircraft architecture.                                       |
| Subsystem Architectures & Interface Modelling | Created all subsystem architecture and interface diagrams illustrating system integration.                       |
| Electrical & Environmental Control System     | Designed and documented the Electrical and Environmental Control System.                                         |
| Avionics Subsystem                            | Designed and documented the avionics subsystem, including navigation, communication, and surveillance systems.   |

This project strengthened my understanding of aircraft systems engineering, requirements development, subsystem integration, and systems architecture.

### Skills Demonstrated

|                                |                                                                                                          |
|--------------------------------|----------------------------------------------------------------------------------------------------------|
| Systems Engineering            | Applied a structured systems engineering process from mission definition to preliminary aircraft design. |
| Mission Analysis               | Defined mission objectives, operational needs, and success criteria.                                     |
| Requirements Analysis          | Translated mission and stakeholder needs into system-level requirements.                                 |
| Concept of Operations (ConOps) | Developed the operational concept describing how the aircraft would fulfil its mission.                  |
| System Architecture            | Produced the high-level aircraft architecture and subsystem integration.                                 |
| Subsystem Design               | Designed the Electrical & Environmental Control System and Avionics subsystem.                           |
| Systems Modelling              | Created subsystem architecture and interface models to support system integration.                       |
| Interface Engineering          | Defined subsystem interfaces to ensure functional integration across the aircraft.                       |
| Engineering Trade Studies      | Evaluated design decisions against operational, certification, cost, and maintainability requirements.   |
| Aircraft Systems Engineering   | Applied systems engineering principles to the conceptual design of a special mission aircraft.           |
| Technical Documentation        | Authored key systems engineering sections of the preliminary design report.                              |
| Team Collaboration             | Collaborated within a multidisciplinary team to develop an integrated aircraft concept.                  |

### Challenges

One of the primary challenges was balancing mission capability with operational, certification, and cost constraints while selecting a suitable aircraft platform. Multiple design trade-offs were required to ensure the proposed solution satisfied the mission requirements without introducing unnecessary complexity or development risk.

Another challenge was integrating multiple aircraft subsystems into a cohesive architecture while maintaining clear interfaces and traceability between mission objectives, system requirements, and subsystem designs. This required close collaboration across the team to ensure all engineering decisions supported the overall mission concept.

### Results

The project produced a complete preliminary systems engineering design for a civilian maritime surveillance aircraft that satisfied the defined mission requirements and operational objectives. The proposed solution integrated mission systems, avionics, electrical, environmental control, communications, and search-and-rescue capabilities into a cohesive aircraft concept, demonstrating the feasibility of adapting an existing platform for New Zealand's maritime operations.

The final design successfully demonstrated the application of systems engineering principles throughout the conceptual design process, from mission definition and requirements analysis to subsystem integration and architecture development. It also reinforced the value of trade studies and multidisciplinary collaboration in developing a practical, cost-effective solution to a complex engineering problem.

**Lessons Learned**

**Images / Diagrams**

[insert system architecture]

[insert conops]

[insert functional block]

[insert lifecycle]

[insert aircraft platform system]

[insert flight and safety]

[insert communication]

[insert mission system]

[insert EECS image]

**Supporting Material**

**Orbital Propagation & Mission Design**

**Overview**

**Project Summary**

|              |  |
|--------------|--|
| Project Type |  |
| Status       |  |
| Institution  |  |
| Discipline   |  |
| Role         |  |
| Focus Areas  |  |



|                  |  |
|------------------|--|
| Key Contribution |  |
| Tools Used       |  |
| Duration         |  |
| Team Size        |  |

### **Objectives**

### **Solution Overview**

### **Engineering Process**

### **My Contributions**

### **Skills Demonstrated**

### **Challenges**

### **Results**

### **Lessons Learned**

### **Images / Diagrams**

### **Supporting Material**

## **Smart Transportation System**

### **Overview**

### **Project Summary**

|                  |  |
|------------------|--|
| Project Type     |  |
| Status           |  |
| Institution      |  |
| Discipline       |  |
| Role             |  |
| Focus Areas      |  |
| Key Contribution |  |
| Tools Used       |  |
| Duration         |  |
| Team Size        |  |

### **Objectives**

### **Solution Overview**

### **Engineering Process**

### **My Contributions**

### **Skills Demonstrated**

### **Challenges**

**Results**

**Lessons Learned**

**Images / Diagrams**

**Supporting Material**

## Wireless Power Transfer

**Overview**

**Project Summary**

|                  |  |
|------------------|--|
| Project Type     |  |
| Status           |  |
| Institution      |  |
| Discipline       |  |
| Role             |  |
| Focus Areas      |  |
| Key Contribution |  |
| Tools Used       |  |
| Duration         |  |
| Team Size        |  |

**Objectives**

**Solution Overview**

**Engineering Process**

**My Contributions**

**Skills Demonstrated**

**Challenges**

**Results**

**Lessons Learned**

**Images / Diagrams**

**Supporting Material**

## Embedded System Design

**Overview**

**Project Summary**

|                  |  |
|------------------|--|
| Project Type     |  |
| Status           |  |
| Institution      |  |
| Discipline       |  |
| Role             |  |
| Focus Areas      |  |
| Key Contribution |  |
| Tools Used       |  |
| Duration         |  |
| Team Size        |  |

## Objectives

## Solution Overview

## Engineering Process

## My Contributions

## Skills Demonstrated

## Challenges

## Results

## Lessons Learned

## Images / Diagrams

## Supporting Material

# Line Adapting Robot

## Overview

## Project Summary

|                  |  |
|------------------|--|
| Project Type     |  |
| Status           |  |
| Institution      |  |
| Discipline       |  |
| Role             |  |
| Focus Areas      |  |
| Key Contribution |  |
| Tools Used       |  |
| Duration         |  |
| Team Size        |  |

## Objectives

## Solution Overview

**Engineering Process**

**My Contributions**

**Technical Skills**

**Challenges**

**Results**

**Lessons Learned**

**Images / Diagrams**

**Tools Used**

## Research

### Orbital Power Hub

**Status: Independent Research (Ongoing)**

The Orbital Power Hub is an independent research initiative exploring a modular orbital infrastructure capable of providing supplemental power, communications, and mission support services to spacecraft in Earth orbit. The concept investigates how shared space-based utilities could improve mission flexibility, reduce spacecraft design constraints, and support the long-term growth of the space economy.

Current research focuses on mission architecture, orbital trade studies, systems engineering, power transmission methods, communications, and operational feasibility. The project represents a long-term research interest that I intend to further develop through postgraduate research and future industry collaboration.

## Achievements

### National Finalist – Falling Walls Lab Aotearoa New Zealand

**2026 | Falling Walls Foundation New Zealand**

Selected as **one of the national finalists** from across New Zealand to present an original aerospace innovation at Falling Walls Lab Aotearoa New Zealand. Presented the **Orbital Power Hub**, a concept proposing orbital infrastructure to enhance spacecraft power and mission capabilities through systems engineering.

## Winning Team – TUC Space Summer School Innovation Challenge

**2026 | Technical University of Crete**

**Awarded 1st place** after collaborating with an international team to develop and present an innovative space mission concept during the inaugural TUC Space Summer School. The challenge required teams to devise a feasible response to a hypothetical asteroid on a collision course with Earth, with only one month before impact. Our solution combined mission design, systems thinking, rapid decision-making, and effective teamwork to deliver the winning proposal among a competitive international cohort.

## Only New Zealand & Oceania Representative – TUC Space Summer School

**2026 | Technical University of Crete**

Selected as the only participant representing New Zealand and the Oceania region at the inaugural TUC Space Summer School in Greece, joining an international cohort to explore spacecraft systems, mission design, space technologies, and astronaut selection.

## Leadership, Communication & Outreach

### International Engineering Collaboration

**2026 | TUC Space Summer School – Greece**

Collaborated with an international, multidisciplinary team to develop an innovative space mission concept under strict time constraints. Worked closely with team members from diverse backgrounds to evaluate ideas, integrate technical solutions, and deliver a successful final presentation.

### Entrepreneurial Innovation Pitch

**2026 | Falling Walls Lab Aotearoa New Zealand**

Presented the **Orbital Power Hub** concept as a national finalist at Falling Walls Lab Aotearoa New Zealand. Delivered a three-minute presentation explaining a complex aerospace systems concept to judges and a multidisciplinary audience, demonstrating concise technical communication under strict time constraints.

### Technical Innovation Presentation

**2026 | ActInSpace New Zealand – SpaceBase Limited**

Worked within a multidisciplinary team to develop and pitch an innovative solution to a real-world space challenge. Combined engineering problem-solving with entrepreneurial thinking by communicating both the technical feasibility and commercial potential of the proposed concept. Supported by CNES and the European Space Agency.

## Engineering Project Exhibition

**2024 | Auckland University of Technology**

Presented my final-year engineering project of a smart transportation system, during the university's Engineering Project Exhibition. Demonstrating the design, development, and outcomes of the project to academic staff, industry representatives, and visitors. Answered technical questions and communicated engineering decisions to audiences with varying levels of technical knowledge.

## Business & Systems Concept Presentation

**2022 | Auckland University of Technology**

Developed and presented a comprehensive Business Model Canvas for SECUREKIWI NZ LTD, a conceptual marine fire alarm company. The project required integrating engineering design with commercial strategy by defining the value proposition, customer segments, partnerships, revenue streams, and operational model, while presenting the concept to academic staff and judges

## Muay Thai Club Event Coordinator

**2021 | Auckland University Muay Thai Club**

Coordinated club events by organising logistics, scheduling activities, and communicating with members to ensure successful event delivery. The role strengthened my leadership, teamwork, organisational, and interpersonal skills while supporting an active university community.

## Contact Information

|          |  |
|----------|--|
| Email    |  |
| Phone    |  |
| LinkedIn |  |
| Website  |  |

## Projects Archive

| Project Name                      | Discipline | Description | Report |
|-----------------------------------|------------|-------------|--------|
| Cosmological Distance & Supernova |            |             |        |
| Home Security System              |            |             |        |
| Rock Paper Scissors (OpenCV)      |            |             |        |
| Analog & Digital Piano            |            |             |        |
| Dual Rail Power Supply            |            |             |        |
| Mobile Solar Powered Trailer      |            |             |        |